Final Project Design Phase

Computer Science - Week 11

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will most likely be using for loops, lists, dataclasses, maybe nested data, and if else statements. While working on this project, I would like to get better at dataclasses and nested data.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the easiest part of the project to be using the if else statements to create different choices throughout the game. I expect the hardest part of this project to be the loops and dataclasses.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The most time will probably be creating the storDy with all the different timelines and

such.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions that may interfere with my work would be being on my phone and talking with friends. To minimize them, I will spend time by myself at the library or some other quiet place and turn my phone off to lock in. I have no further questions.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

story of the game……

“What is your next choice?”

Choice A	Choice B
Choice C	Choice D

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	I will create a story game where a player will explore a dungeon and has to choose paths to get loot and escape.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	player_name (str): Name of your player. hours_left (int): How much time is left in the dungeon before you need to leave. dungeon_room (str): Type of room the player is in. money (int): How much money they have on them. Can unlock different things
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Item: name (int): name of item description (str): what the item does value (int): how much the item is worth

Routes (minimum 7) The names of your route functions	1. start_game 2. choose_room 3. leave_dungeon 4. restart 5. finale 6. view_items 7. time_left
Pages (minimum 5) A brief description of the pages in your site	1. Dungeon rooms: where you make your choices. 2. Inventory of items: page to look at your items. 3. Victory: final screen when you win. 4. Fail: final screen when you lose. 5. Starting screen: Where you can start the game.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	name (textbox): player enters their name. cave (selectbox): player chooses which cave to enter. take_item (checkbox): player chooses to pick item up or not.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. In choose_room, if choice == cave 1 / elif choice == cave 2, etc. 2. In time_left If hours <= 0, game over 3. In start game, If name == “”, return “please enter name”
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	In the finale, for items in inventory, if item.value > x, allow for power up.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

i need to create a story game where choices will determine different outputs. what are some if statement formats i can use

Listing if-statement formats for the story game.

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Here are the if statement formats that work well for a story game. They fall into a few groups, and the last section shows how to combine them.

1. Basic if / else (two outcomes)

Use it for a simple either-or choice, like a checkbox.

python

```
if take_item:
    state.inventory.append(Item("Matchbook", "Dry and ready to strike.", 2))
else:
    state.courage = state.courage - 1
```

2. if / elif / else chain (three or more outcomes)

Use it when the player picks one option from a SelectBox. Python checks the conditions top to bottom and runs only the first one that's true.

python

```
if door == "Library":
    page_text = "Dusty books line the walls."
elif door == "Kitchen":
    page_text = "Something is still simmering on the stove."
else:
    page_text = "The cellar stairs creak under you."
```

3. Comparing numbers (thresholds)

Use it for stats like courage, hours, or health. The comparison operators are <, >, <=,

>=, ==, and !=.

python

```
if state.courage >= 3:
    ending = "You face the ghost and walk past."
else:
    ending = "You turn and run."
```

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.